

The Outside Network Area

The Living Map · TSY14 Technical Challenge · Design brief and specification

Written on **03/10/2026** · Official Phase 1 submission deadline: **05/10/2026** · Companion to “Phase 1 Project Status & Technical Handover”

Who this is for

- The person who owns the Outside Network Area in the team (the AI assumed one of four roles: Writer, Beacon, **Outside Network**, Executor) — and any new AI agent helping them.
- It explains what the area is, what the organizers require, what has been designed so far, and what is still open.
- **Nothing in the Outside Network Area exists in code yet.** The only related code is the Writer side of the simulation, which produces the log this area will receive.
- Labels: **[CONFIRMED]** official document or user · **[PROPOSED]** suggested by the AI, not approved · **[ASSUMPTION]** invented to make the design work · **[OPEN]** no decision yet · **[DESIGN ONLY]** not implemented · **[INTERPRETATION]** the AI’s reading of the documents.

1. What the Outside Network Area is

Inside the disaster zone there is **no GPS and no network**. Outside it, **GPS, satellite time and communication are available**. The Outside Network Area is the **bridge that connects the two worlds**. The organizers say that teams design its interior and that it is **defined by its roles, not by its parts** (slide 5).

[CONFIRMED]

1.1 What the official documents require

Requirement (source)	Meaning
Implement a functional Outside Network Area (instructions, slide 6)	It must really work in the demo. [INTERPRETATION] A simulated satellite link is probably acceptable in Phase 1, but the area itself must run.
Route all inside↔outside communication through it	The Writer, the Executor and the command post never talk to each other directly.
No direct link between the robots and the command post	Even for convenience or debugging, do not connect a robot straight to the command post.
Four roles: RECEIVE · TRANSLATE · CARRY · BRIEF (slide 5)	See section 2.
Technical focus: command-post connection · satellite / wireless communication · data reception and forwarding (cahier des charges p.2)	The link to the command post can be wireless or satellite.
Frame translation: private robot coordinates → real-world GPS coordinates	See section 5.
Display a live map at the command post (goal list, p.1)	See the open question on “live” in section 9.

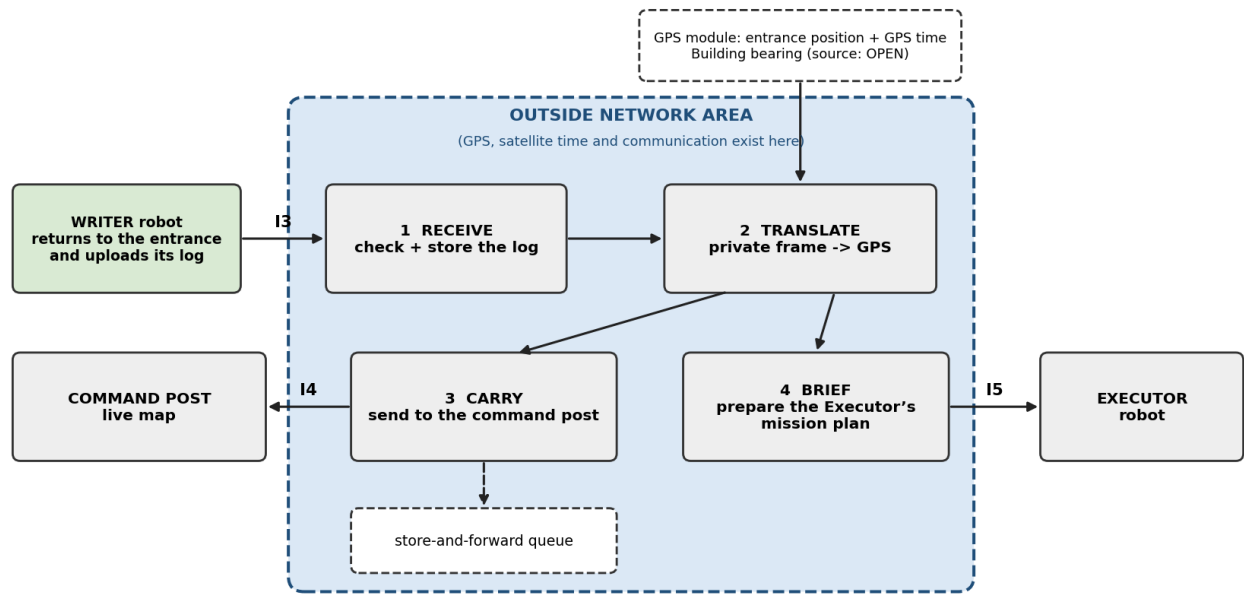
1.2 Where it sits in the official chain

Writer explores → events detected & beacons deposited → **Outside Network receives & translates data → wireless / satellite communication → command post** → Executor receives mission → Executor enters & navigates using beacons.

1.3 Points at stake in Phase 1

Outside network area: 5 · Frame translation: 5 · Technical solution diagrams: 5 (shared) · Failure cases: 3 (shared) · Implementation plan: 3 (shared). The Phase 1 deliverables also include the simulation demo (8 points), which should show this area working. The Final Phase adds a physical prototype (15 points).

2. The four roles in detail



$I3$ = Writer log · $I4$ = GPS-referenced data for the map · $I5$ = briefing (mission, beacon chain, clock sync, trust rules)
 Green = exists in the Python simulation (the Writer side). Grey = designed only: nothing in the Outside Network exists in code yet.

Figure 1 — Internal structure of the area (proposed). The Writer side already exists in the simulation; everything else is a design.

Role	Input	What it does (proposed)	Output
1 RECEIVE	The Writer's log ($I3$), sent when the Writer is back at the entrance	Accepts the upload; checks that required fields are present and the data is complete (a checksum is suggested); stores the raw log; ignores a duplicate of a log already received (same start time). If the Writer never returns, reports "no log".	A validated log, in the Writer's private frame
2 TRANSLATE	Validated log + GPS fix of the entrance + building bearing	Converts every beacon, event and path point from the private frame to latitude/longitude (section 5). Attaches an uncertainty estimate that grows with distance.	GPS-referenced events, beacons and path
3 CARRY	GPS-referenced data	Sends it to the command post over the chosen link. If the link is down, keeps it in a store-and-forward queue and resends later.	Data ($I4$) that feeds the live map
4 BRIEF	The mission chosen at the command post + the translated data	Builds the Executor's briefing: mission, ordered beacon chain from the entrance to the target, known hazards, clock synchronisation, trust rules. Delivers it to the Executor before it enters.	Briefing ($I5$)

3. How it works in time (proposed sequence)

Step	What happens	Who / note
0 · Before entry	The gateway takes a GPS fix of the entrance and GPS time, records the building bearing, and sets the Writer's clock.	Clock sync is an [ASSUMPTION] (needed for beacon timestamps and aging).
1 · Writer inside	The Writer explores alone. No contact with the outside — no network exists.	Official condition.
2 · Writer returns	At the entrance the Writer uploads its log (I3). In the simulation the upload only succeeds within 1.5 m of the entrance, up to 3 tries.	RECEIVE. The 1.5 m is a simulation [ASSUMPTION] .
3 · Translate	The log is converted to GPS coordinates.	TRANSLATE.
4 · Carry	The data is sent to the command post; the live map updates.	CARRY. The map therefore updates when the log arrives, not while the Writer is exploring.
5 · Mission	The command post chooses the mission (extinguish, extract, seal or retrieve).	Who chooses — a human operator or a rule — is [OPEN] .
6 · Brief	The briefing (I5) is prepared and delivered to the Executor; its clock is set.	BRIEF. How it is delivered (Wi-Fi at the entrance, cable...) is [OPEN] .
7 · Executor inside	The Executor enters and navigates by the beacons. Again no contact with the outside.	Executor teammate.

4. Interfaces

4.1 Input: the Writer's log (I3) **[DRAFT]** **[IMPLEMENTED IN SIM]** (Writer side)

The Writer simulation already produces this JSON (structure shown, values are placeholders). All coordinates are in the Writer's **private frame**: origin = entrance, x = direction of entry, y = left.

```
{ "robot": "writer", "frame": "private", "origin": "entrance", "start_time": <epoch s>, "end_time": <epoch s>,
  "status": "returned", "return_reason": "...",
  "path": [[x, y, heading_rad, t], ...],
  "beacons": [{"id", "type", "dir_deg", "dist_m", "t", "ver", ["exits"], "x", "y", "severity", "score", "confidence"}, ...],
  "decisions": [{"t", "type", "action", "score", "reason", "x", "y"}, ...],
  "stats": {"path_length_m", "heading_corrections", "drop_failures", "stock_left", "battery_end", "stuck_events"} }
```

Beacon type is HAZARD, JUNCTION, HOLE or WAYPOINT. dir_deg is 0–359° in the private frame (0 = direction of entry, counter-clockwise). If the Writer is lost, **no log exists**. The log has not been agreed with the Outside Network owner, who may change the fields (**[OPEN]**).

4.2 Output to the command post (I4) **[PROPOSED]** **[DESIGN ONLY]**

```
{ "entrance": {"lat", "lon", "bearing_deg"}, "generated_at": <epoch s>, "writer_status": "returned | no_log",
  "events": [{"beacon_id", "type", "lat", "lon", "severity", "t", "uncertainty_m"}, ...],
  "path": [[lat, lon], ...], "warnings": ["gps_missing", ...] }
```

4.3 Output to the Executor (I5, the briefing) **[PROPOSED]** **[DESIGN ONLY]**

```
{ "mission": "EXTINGUISH | EXTRACT | SEAL | RETRIEVE", "target_beacon_id": <id>,
  "entrance_pose": {"heading_private_deg": 0},
  "chain": [{"order", "beacon_id", "type", "dir_to_next_deg", "dist_to_next_m"}, ...],
  "hazards": [...], "clock_sync_epoch": <s>,
  "trust_rules": {fresh, aging, stale, suspect thresholds – values OPEN} }
```

[OPEN DESIGN ISSUE] Who builds the beacon chain? When the Writer drops a beacon, it does not know where the *next* beacon will be. So each beacon holds information about its own event, and the true beacon-to-beacon directions and distances only become known from the complete log. The Outside Network (BRIEF) can compute the ordered chain from the log and give it to the Executor in the briefing. This is how the current design works, but it needs to be agreed with the Beacon and Executor teammates (see the handover PDF, section 8.2).

5. Frame translation: private coordinates → GPS

5.1 The idea [PROPOSED] [DESIGN ONLY]

The Writer has no GPS, so it measures everything in its own frame. The gateway knows (a) the **GPS position of the entrance** (lat_0 , lon_0) and (b) the **compass bearing b of the +x direction**, measured clockwise from north. For a point (x , y) in the private frame:

```
East  = x·sin(b) - y·cos(b)
North = x·cos(b) + y·sin(b)
lat   = lat0 + North / 111 320
lon   = lon0 + East / (111 320 · cos lat0)
```

111 320 is roughly the number of metres in one degree of latitude. This flat-earth approximation is adequate over tens of metres. **[ASSUMPTION]** The formulas are proposed, not yet coded or checked against a real GPS.

5.2 Worked example (illustrative numbers only)

Made-up entrance at $lat_0 = 36.8000^\circ$, $lon_0 = 10.1800^\circ$; entry bearing $b = 60^\circ$:

Beacon (private frame)	East / North from the entrance	Result (lat, lon)
(1.0 m, 0.0 m)	0.87 m / 0.50 m	36.800004, 10.180010
HAZARD at (8.0 m, 2.0 m)	5.93 m / 5.73 m	36.800051, 10.180067
HOLE at (14.0 m, -3.0 m)	13.62 m / 4.40 m	36.800040, 10.180153

5.3 Where the building bearing could come from [OPEN]

Option	Idea	Trade-off
A. Compass reading	Read a magnetometer outside, aligned with the entry direction	Cheap; distorted by metal and electronics — keep it away from the robot.
B. Two GPS points	Take GPS fixes at the entrance and a second point along the entry line	No compass; accuracy limited by the GPS error over a short baseline.
C. Map / satellite image	The operator reads the bearing from a map	Simple and robust for a demo; manual step.
D. Stakes on the ground	Mark the entry direction physically and measure it	Very controlled; useful for the demo arena.

5.4 Error budget — to state honestly in the report

- A bearing error of 3° moves a point 10 m away about **0.5 m sideways** ($10 \text{ m} \times \sin 3^\circ$). The error grows with distance.
- The Writer's dead-reckoning drift also grows with distance, so far beacons have less certain GPS positions. In the simulation the final position error was 0.02–0.08 m, but those use **simulated** sensors and prove nothing about real hardware.
- Consumer GPS is typically accurate to several metres (check the datasheet of the chosen module). For the command-post map this may dominate; for the Executor, local beacon-to-beacon guidance stays accurate, so translation matters most for the map and for reporting hazard positions.
- Recommended in the report: show the uncertainty on the map and say which parts are measured and which are assumed.

6. Design options

Level	Idea	Advantages	Limits
Simple	A laptop runs a Python program (receive, translate, queue, brief). GPS from a USB/phone GPS or fixed test coordinates. The "satellite" link is a Wi-Fi or local-network simulation. Map shown in a browser.	Fast; enough for the Phase 1 simulation; easy to test failure cases.	Weak as a physical prototype.
Intermediate	Raspberry Pi gateway with a GPS module at the entrance, a message broker (MQTT) and a web map at the command post; a long-range radio or a cellular link between gateway and command post.	Realistic; a strong story for the judges; close to the final prototype.	More hardware to build and debug.
Ambitious	A real satellite modem or long-range link with full store-and-forward.	Most impressive.	Costly and risky; not needed for Phase 1.

[PROPOSED] Recommendation: simulate the **simple** level for Phase 1, and **describe the intermediate level** as the hardware plan for the final phase. This is a suggestion; the owner of this area decides. Hardware

candidates for the final phase (none chosen): a small computer as gateway, a GPS module, a local link to receive the Writer's log at the entrance (the UNO Q is believed to have Wi-Fi — **[ASSUMPTION]**, to verify), a long-range link to the command post, a PC with a browser map, and power for the gateway outdoors.

7. Phase 1 simulation plan

Goal: a small program that behaves like the area, takes the Writer's log as input, and shows each role working. Everything below is **[NOT STARTED]**.

- **Four functions**, one per role: `receive(log)`, `translate(log, gps0, bearing)`, `carry(data, link)` with a `queue`, `brief(data, mission)`.
- **Inputs**: the Writer simulation's `writer_log.json`. The Writer simulation is not fully reliable yet, so a hand-made log in the same format is a good starting point.
- **Outputs**: a GPS-referenced data file (I4), a briefing file (I5), and a simple map picture or web page showing events and path.
- **Link**: a software object that can be switched up and down, to demonstrate store-and-forward.

Test	Input	Expected result	Status
T1 Normal	Valid log	GPS file, map and briefing are produced	[NOT STARTED]
T2 Link down	Valid log; link off, then on	Data queued, then delivered when the link returns	[NOT STARTED]
T3 No log	Writer lost, nothing received	Status "no_log"; briefing marked "no inherited data"; the Executor must rely on beacons only	[NOT STARTED]
T4 Bad log	Truncated or corrupted file	Rejected with a clear message; re-upload requested	[NOT STARTED]
T5 No GPS fix	Valid log; no GPS position	Map in the private frame with a visible warning	[NOT STARTED]
T6 Duplicate	Same log uploaded twice	Processed once	[NOT STARTED]

8. Failure cases of this area [DESIGN ONLY]

What fails	Consequence	Planned response
Link to the command post goes down	The map is not updated	Store-and-forward queue; resend; show "last update" time
The Writer is lost (no log)	The area has no data	Report "no log"; the Executor is briefed with the mission only and relies on beacons
GPS fix missing or poor	Positions cannot be put on a real map	Show the private frame and a warning; show uncertainty when the fix is poor
Log corrupted or incomplete	Wrong or missing beacons on the map	Validate fields and checksum; reject and ask for a re-upload
Wrong building bearing	The whole map is rotated	Check against known landmarks; display the bearing used so it can be questioned
Clock not synchronised	Beacon aging is wrong	Synchronise from GPS time before entry; log the sync time
Briefing not delivered	Executor enters without its mission	Do not release the Executor until delivery is acknowledged

9. Open decisions

"Responsible" assumes the four-role split; the real names are unknown. No decision below has been made.

Item	Current status	Decision needed	Responsible
Building bearing source	Four options in 5.3	Choose one for the demo and the final prototype	Outside Network owner
Who builds the beacon chain	Outside Network (BRIEF) in the current design	Agree with Beacon and Executor teammates	Beacon + Outside Network + Executor
Log format (I3)	Draft produced by the Writer sim	Confirm or change the fields	Outside Network owner + Writer owner
Briefing format (I5) and delivery method	Sketch in 4.3; delivery unknown	Fix fields; choose how the Executor receives it	Outside Network + Executor

Item	Current status	Decision needed	Responsible
Trust / aging thresholds	Slide examples 2/15/40 min; AI proposal 5/30 min; not coded	Choose values and who applies them	Beacon + Executor
Who chooses the mission	Unknown	Human operator at the command post, or an automatic rule	Whole team
“Live” map	Updates only when the Writer’s log arrives	Is that acceptable as “live”? Ask the organizers if unsure	Whole team (aess@ieee.tn, ras@ieee.tn)
Link technology	Wireless or satellite allowed; none chosen	Choose what is simulated now and what is built later	Outside Network owner
Clock synchronisation	GPS time before entry (assumption)	Who sets the clocks and how	Outside Network owner
What is real in the final prototype	Nothing physical yet	Decide the hardware list (section 6)	Outside Network owner

10. What this area must deliver, and next steps

	Task	Why
☐	Confirm the log format (I3) with the Writer owner, and the briefing format (I5) with the Executor teammate	Everything else depends on the interfaces
☐	Decide the building-bearing source and the simulated link	Needed for translation and the demo
☐	Write the four functions and run tests T1-T6 on a sample log	Simulation demo (8) and Outside network area (5)
☐	Write the frame-translation note with the worked example and the error budget	Frame translation (5)
☐	Draw the final architecture and data-flow diagrams of the area	Technical solution diagrams (5)
☐	Write the failure cases (section 8) and record T2-T6 results	Failure cases (3)
☐	List the final-phase hardware, a timeline to 01/12/2026, and the risks	Implementation plan (3); physical prototype (15) later
☐	Write the “Outside Network and Executor” part of the 6-page report (about 0.7 pages shared with the Executor)	Technical report (2)

Not yet done by anyone: opening the two links inside the official cahier des charges — the “System Architecture Diagram” resource (a Google Drive file) and the submission form. The architecture diagram resource may show what the organizers expect for this area, so open it first. Links are listed in section 2.2 of the handover PDF.